Isotopes

Exam code: 0620 & 0971

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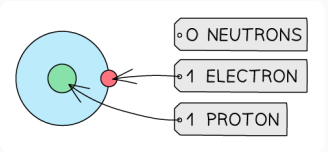
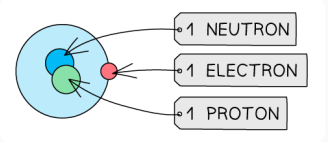
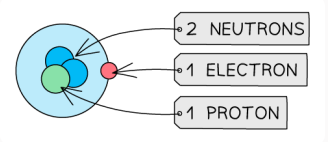
Defining isotopes

- Isotopes are different atoms of the **same element** that contain the same number of **protons** but a different number of **neutrons**
- The symbol for an isotope is the chemical symbol (or word) followed by a dash and then the mass number
- So C-14 (or carbon-14) is the isotope of carbon which contains 6 protons, 6 electrons and 14 - 6 = 8 neutrons
  - It can also be written as ^{14}C or $^{14}_{6}C$

Table to show the structures of isotopes of hydrogen

Isotope	Atomic Structure	Symbol
Hydrogen - 1		$^{1}_{1}H$
Hydrogen - 2		$^{2}_{1}H$
Hydrogen - 3		$^{3}_{1}H$

Why isotopes share properties

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- This is because they have the same number of electrons in their outer shells, and this is what determines their chemistry
- The difference between isotopes is the neutrons which are neutral particles within the nucleus and add mass only
- The difference in mass affects the physical properties, such as density, boiling point and melting point
- Isotopes are identical in appearance, so a sample of C-14 would look no different from C-12

Calculating relative atomic mass

Extended tier only

Relative atomic mass

- Atoms are so tiny that we cannot really compare their masses in conventional units such as kilograms or grams, so a unit called the **relative atomic mass** (A_r) is used
- The **relative atomic mass** unit is equal to 1/12th the mass of a carbon-12 atom
- All other elements are measured by comparison to the mass of a carbon-12 atom and since these are ratios, the relative atomic mass has no units
- For example, hydrogen has a relative atomic mass of 1, meaning that 12 atoms of hydrogen would have exactly the same mass as 1 atom of carbon

How do I calculate relative atomic mass?

- The relative atomic mass of each element is calculated from the **mass number** and **relative abundances** of all the isotopes of a particular element
- The equation below is used where the top line of the equation can be extended to include the number of different isotopes of a particular element present

$$A_r = \frac{(\text{\% of isotope A x mass of isotope A}) + (\text{\% of isotope B x mass of isotope}}{100}$$

- So, if there were 3 isotopes present then the equation would read:

$$\frac{(\text{\% of isotope A x mass of isotope A}) + (\text{\% of isotope B x mass of isotope B}) +}{100}$$

Worked Example

The table shows information about the isotopes in a sample of rubidium

Isotope	Mass Number	Percentage abundance

Use information from the table to calculate the relative atomic mass of this sample of rubidium.

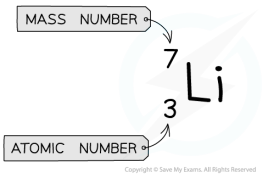
Give your answer to one decimal place.

Answer:

- Relative atomic mass = $\frac{(72 \times 85) + (28 \times 87)}{100}$
- Relative Atomic Mass = **85.6**

Is mass number and relative atomic mass the same thing?

- On the Periodic Table provided in your exam you will see that lithium has a relative atomic mass of **7**
- Although it seems that this is the same as the mass number, they are not the same thing because the relative atomic mass is a **rounded** number
- Relative atomic mass takes into account the existence of isotopes when calculating the mass
- Relative atomic mass is an **average mass** of all the isotopes of that element
- For simplicity **relative atomic masses** are often shown to the nearest whole number



The relative atomic mass of lithium to two decimal places is 6.94 when rounded to the nearest whole number, the RAM is 7, which is the same as the mass number shown on this isotope of lithium

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